

📘 AI/ML Internship – Day 24 Notes & Tasks

🟡 Module 4: NumPy (Random Module & Real Dataset Simulation)

📌 Part 1: Why Random Module is Important?

👉 In AI/ML:

- Random data is used for:
 - Testing
 - Simulation
 - Model training
 - Dataset creation

👉 Machine Learning heavily uses randomness.

📌 Part 2: Theory Answers (IMPORTANT)

◆ 1. What is NumPy Random Module?

👉 NumPy random module is used to generate random numbers and random datasets.

◆ 2. Why Random Data is Important?

👉 It helps:

- Simulate real-world data
 - Test ML models
 - Create sample datasets
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◆ 3. What is Random Seed?

👉 Seed ensures the same random output every time.

👉 Useful for reproducibility.

◆ 4. Difference Between randint() and rand()

Function Purpose

randint() Random integers

rand() Random decimal values

◆ 5. What is Dataset Simulation?

👉 Creating artificial data similar to real-world data.

✂ Part 3: Random Integers

```
import numpy as np
```

```
print(np.random.randint(1, 10))
```

◆ Multiple Random Integers

```
print(np.random.randint(1, 100, size=5))
```

✂ Part 4: Random Decimal Numbers

```
print(np.random.rand(3))
```

◆ 2D Random Array

```
print(np.random.rand(2, 3))
```

✂ Part 5: Random Seed

◆ Why Important?

👉 Same results every time.

```
np.random.seed(42)
```

```
print(np.random.randint(1, 10, size=5))
```

📌 Part 6: Random Choice

```
colors = ["red", "blue", "green"]
```

```
print(np.random.choice(colors))
```

◆ Multiple Choices

```
print(np.random.choice(colors, size=5))
```

📌 Part 7: Shuffle Data

```
arr = np.array([1, 2, 3, 4, 5])
```

```
np.random.shuffle(arr)
```

```
print(arr)
```

📌 Part 8: Simulating Student Dataset

```
marks = np.random.randint(40, 100, size=(5, 3))
```

```
print(marks)
```

👉 Example Meaning:

- 5 students
 - 3 subjects
-

📌 Part 9: Simulating Image Data

```
image = np.random.randint(0, 256, size=(3, 3))
```

```
print(image)
```

👉 Represents grayscale image pixels.

📌 Part 10: Real-World Thinking

👉 In AI:

- Random weights initialize neural networks
- Shuffle prevents model bias
- Random sampling improves training

👉 Used in:

- Machine Learning
 - Deep Learning
 - Data Science
-

📌 Part 11: Visualization of Simulated Dataset

Student Marks Dataset

```
[[85 70 90]  
 [60 75 80]  
 [95 88 92]  
 [55 67 72]  
 [78 81 84]]
```

Rows → Students

Columns → Subjects

Task 1: Theory (Write Answers)

1. What is NumPy random module?
2. Why random data is important?
3. What is seed?
4. Difference between randint and rand
5. What is dataset simulation?

Task 2: Random Integer Practice

1. Generate random integer
2. Generate array of integers
3. Create random matrix

Task 3: Random Decimal Practice

1. Generate decimal values
2. Create 2D decimal matrix

Task 4: Random Choice

1. Pick random color
2. Generate multiple choices
3. Simulate random names

Task 5: Shuffle

1. Shuffle array
 2. Shuffle student IDs
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Task 6: Dataset Simulation

1. Create marks dataset
2. Create attendance dataset
3. Create random image matrix